

Between-sample quality control and exploratory diagnostics

Groups

Reference: WT. Comparison: KJ.

Purpose

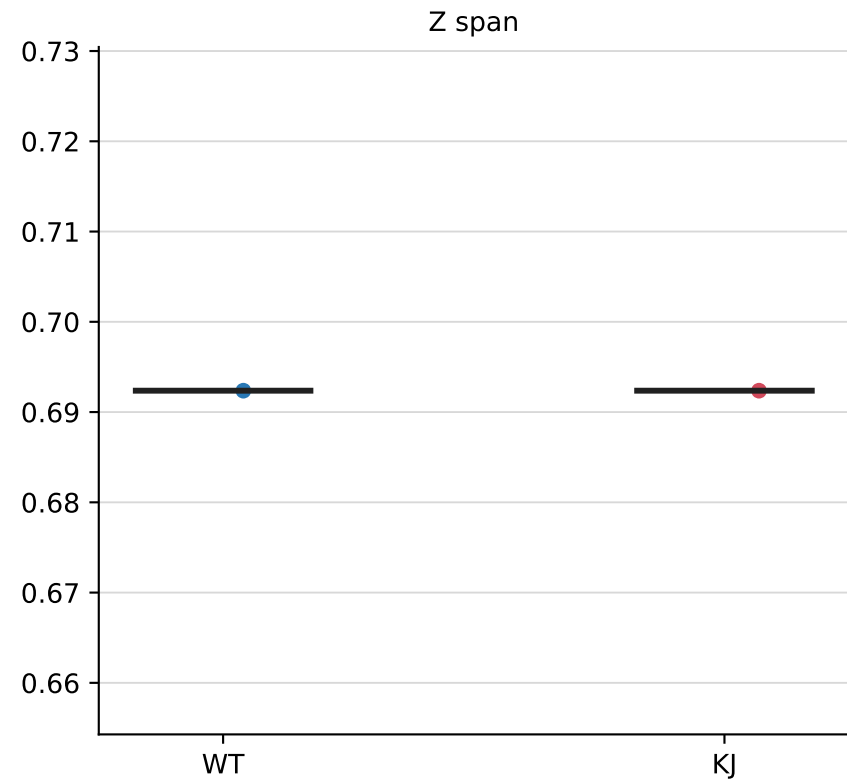
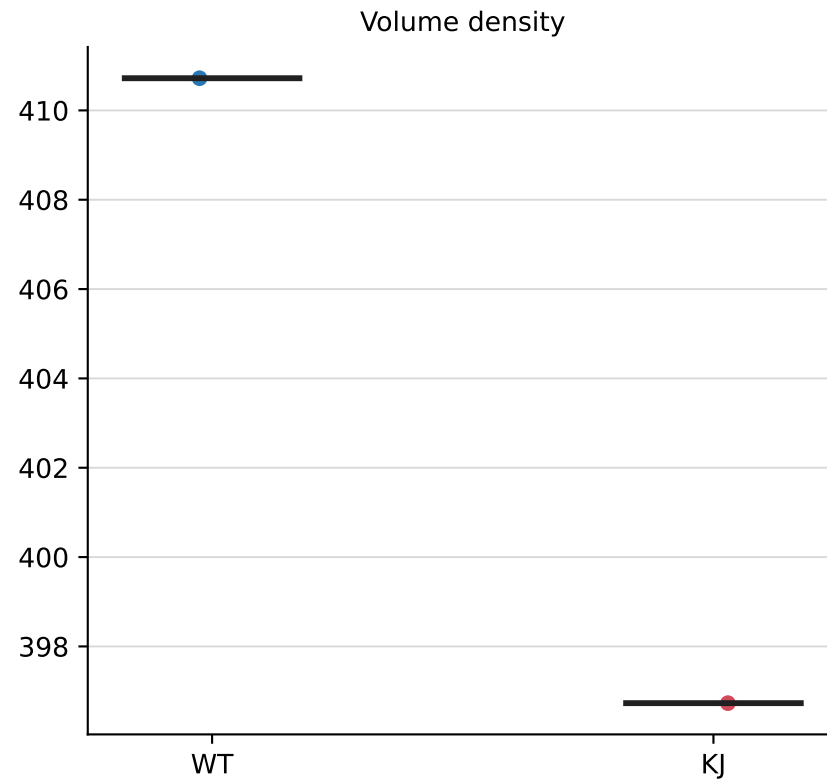
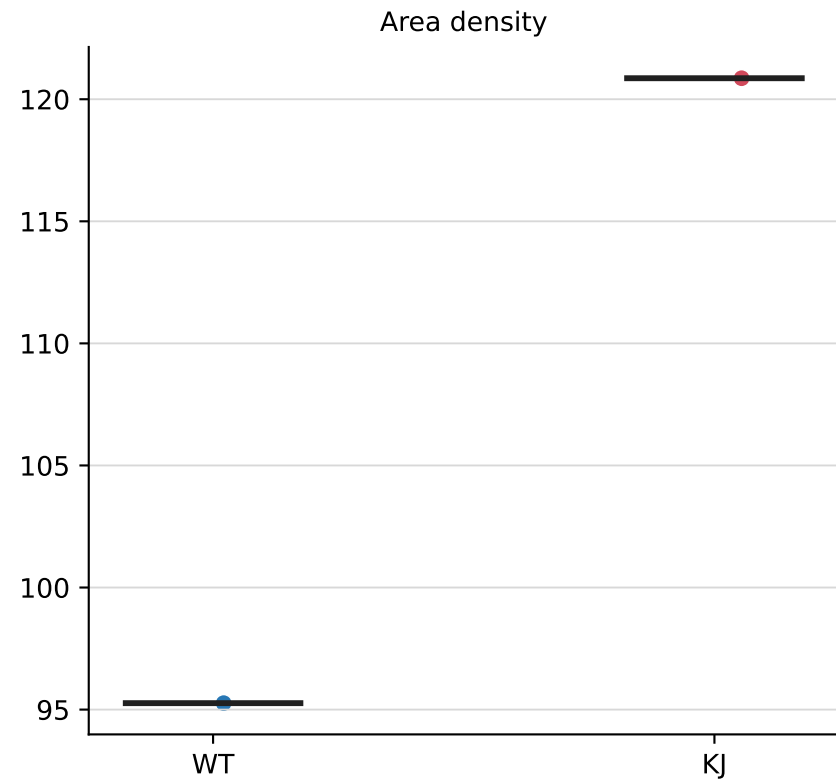
Use this package to judge acquisition comparability, normalization, tracking behavior, and warning frequencies.

Primary results

Biological conclusions belong in ../01_biological_results.

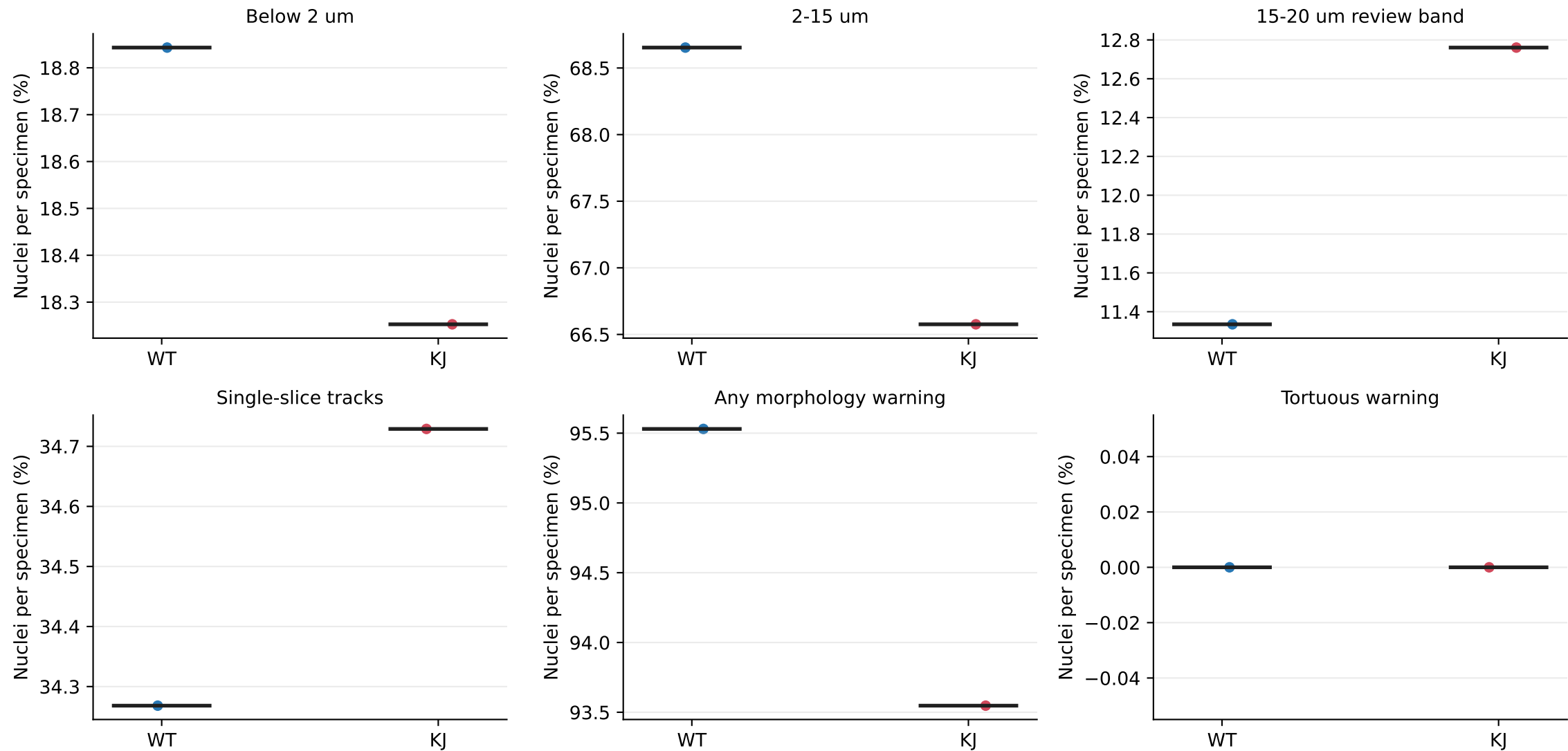
Exploratory count and tracking endpoints

Each point is one specimen; black bars show medians



Per-specimen morphology and tracking proportions

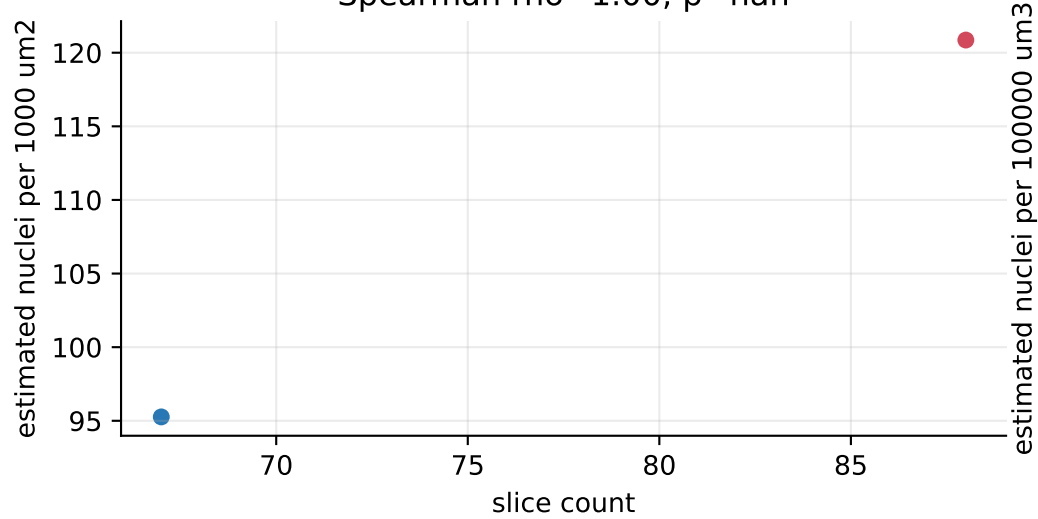
Categories annotate technical-valid nuclei; they are not rejection populations



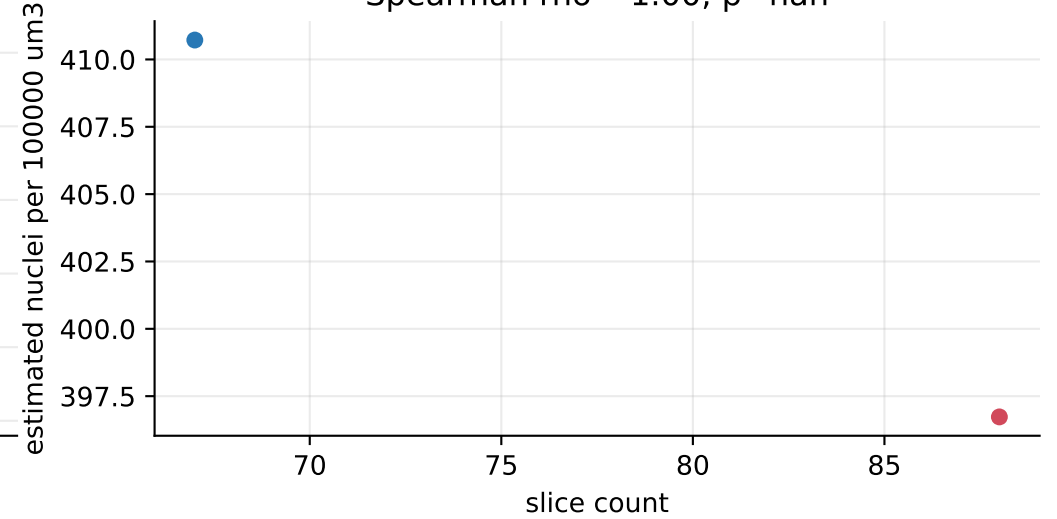
Count and acquisition-depth diagnostics

Count endpoints remain exploratory

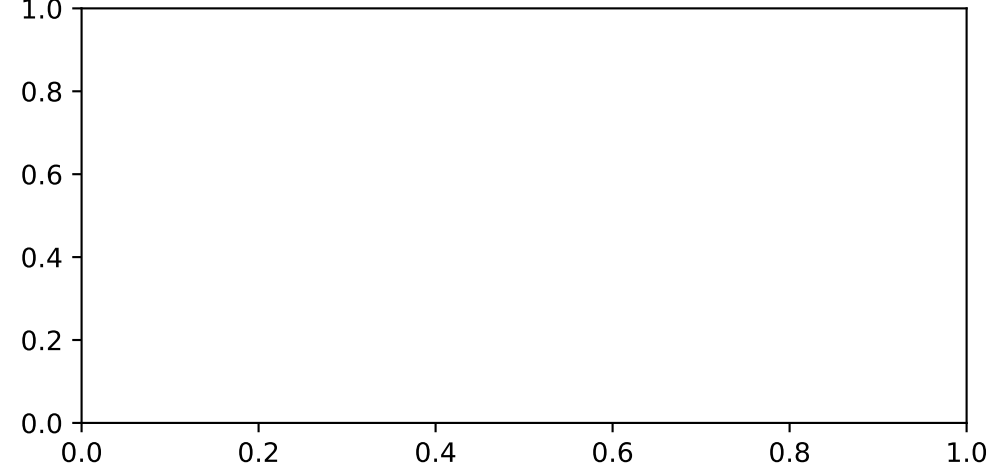
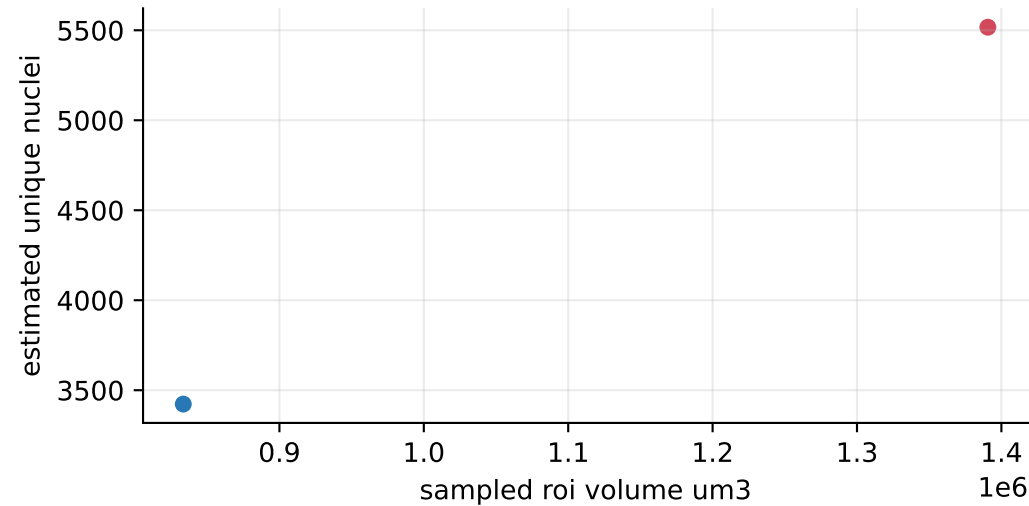
Area density versus stack depth
Spearman rho=1.00, p=nan



Volume density versus stack depth
Spearman rho=-1.00, p=nan



Raw estimated nuclei versus sampled volume
Spearman rho=1.00, p=nan



● WT ● KJ

Specimen-level statistical summary

Each value is calculated across biological specimens, not pooled nuclei

Measure	WT median	KJ median	Median change (%)	Effect size Cliff's delta	Median test FDR q	Rank test FDR q	Mean test FDR q
Estimated nuclei per 1,000 um2	95.3	121	+26.9%	+nan	nan	nan	nan
Estimated nuclei per 100,000 um3	411	397	-3.4%	+nan	nan	nan	nan
Specimen median Z span (um)	0.692	0.692	-0.0%	+nan	nan	nan	nan

Medians: the typical specimen value in each group. Median change: (KJ median - WT median) / WT median x 100; positive means higher in KJ. Cliff's delta: direction and separation, from -1 to +1; positive means KJ specimens tend to have higher values. FDR q: p-value adjusted for testing multiple measurements; q < 0.05 is evidence of a group difference. Median test: permutation comparison of specimen medians (primary). Rank test: Mann-Whitney comparison of specimen ordering. Mean test: Welch comparison of specimen means.

How to use the quality-control package

Not the primary biological result

These panels assess acquisition depth, count normalization, tracking span, morphology-warning frequencies, and processing provenance. They should not replace the biological morphology report.

Count endpoints

Area- and volume-normalized counts remain exploratory when they retain an association with slice count or sampled volume. The fixed-depth sensitivity panel helps assess this dependence.

Morphology warnings

Warnings annotate unusual but technical-valid nuclei. They are not rejection populations and must not be used to force either group toward reference morphology.

Tracking Z span

Z span is sensitive to acquisition spacing and cross-slice linking. A single-slice nucleus can still be biologically genuine.